

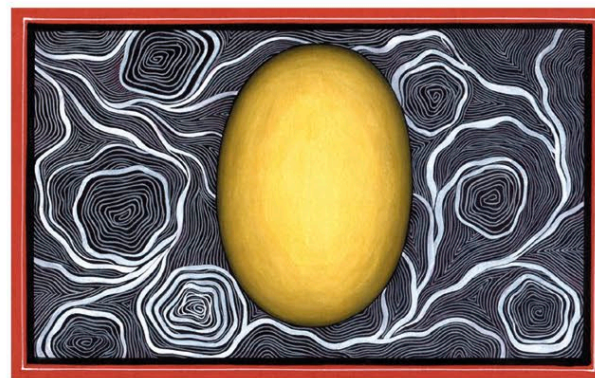
THE NATURE OF THE UNIVERSE

COSMOLOGY—THE STUDY OF THE UNIVERSE AS A WHOLE—IS A FIELD OF ASTRONOMY THAT SEEKS TO ANSWER FUNDAMENTAL QUESTIONS CONCERNING THE SIZE, AGE, AND STRUCTURE OF THE UNIVERSE.

Philosophers and astronomers have been grappling with such questions for thousands of years, with mixed success. The answer to one of the biggest—whether the Universe is finite or infinite in extent—is still not known for certain (although an infinite Universe seems more likely). Other fundamental questions about the nature of the Universe for which answers are now known include how and when the Universe began, whether it has any center or edges, and whether it encompasses more than just our galaxy.



Georges Lemaître



Modern depiction of Hiranagarbha

c. 1500–1200 BCE

Cosmic Egg

Hindu text the *Rigveda* contains a hymn that describes the Universe as originating from a cosmic golden egg or womb known as Hiranagarbha. This floated in darkness before breaking apart to give rise to Earth, the heavens (space), and underworlds.

► 4th Century BCE

Aristotle's Earth-centered Universe

The Greek philosopher Aristotle proposes a Universe that is finite in extent, but infinite in time and has a stationary Earth at its center. Aristotle outlined a complex system containing 55 spheres, the last of which marked out the "edge" of the Universe.

◀ 1931

Primeval atom

Belgian astronomer and priest Georges Lemaître proposes his "hypothesis of the primeval atom." This suggests that the Universe has expanded from an initial extremely hot, dense state. His model also provides a solution to Olbers' paradox.

◀ 1920s

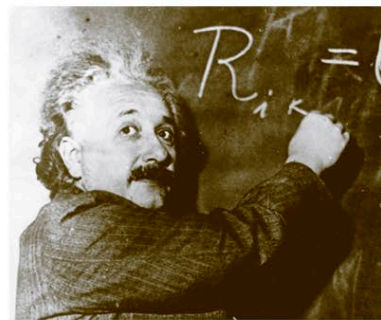
Expanding Universe

American astronomer Edwin Hubble proves that galaxies exist outside our own and observes that distant galaxies are moving away from us at a rate proportional to their distance. Other astronomers conclude that the whole Universe must be expanding.

◀ 1915

General Theory of Relativity

Einstein publishes his General Theory of Relativity, viewed today as the best account of how gravity works on cosmic scales. It proposes that concentrations of mass warp spacetime. He also devises equations that define various possible universes.



Albert Einstein

► 1948

The first elements

Russian-American physicist George Gamow and others work out how—starting with just subatomic particles (in this case protons and neutrons)—the nuclei of different light elements could have formed soon after the start of a very hot, dense, but rapidly expanding Universe.

► 1949

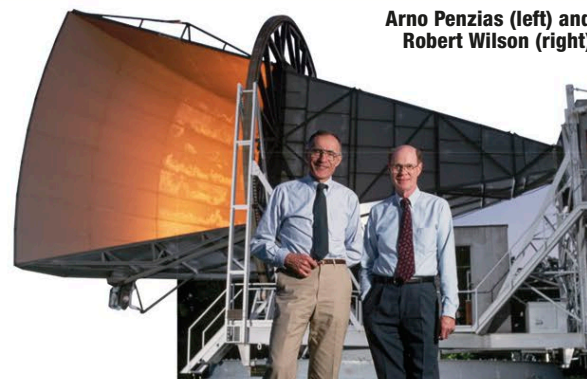
Hoyle coins the term "Big Bang"

British astronomer Fred Hoyle coins the term "Big Bang" for theories that propose the Universe expanded from an exceedingly hot, dense state at a specific moment in the past. The term becomes popular, although Hoyle himself believes in a different theory.

► 1965

Cosmic Microwave Background Radiation

Arno Penzias and Robert Wilson, astronomers at Bell Labs in New Jersey, discover the Cosmic Microwave Background Radiation (CMBR)—a faint glow of radiation coming from everywhere in the sky. It comes to be realized that this is a relic of the Big Bang.



Arno Penzias (left) and Robert Wilson (right)